

The empirical recurrence for OEIS A295203: recovering a conjecture that is not written down

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Abstract

OEIS A295203 records “Empirical recurrence of order 85”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 255$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 85, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 85 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295203 is “Number of nX6 0..1 arrays with each 1 horizontally or vertically adjacent to 2 or 4 1s.”. It has offset 1 and begins

1, 9, 53, 284, 1597, 8241, 40924, 205338, ...

The entry states:

Empirical recurrence of order 85 (see link above)

As of the “Last modified” line on the live entry (Nov 16 2017 21:58:18, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 255$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 255$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 510$ exact terms of the model returns order 85 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{85} c_i a(n-i),$$

c_1	7	c_{23}	−3603466	c_{45}	510985478	c_{67}	3579078
c_2	−15	c_{24}	−6033535	c_{46}	466117789	c_{68}	5523787
c_3	25	c_{25}	−5859961	c_{47}	285366263	c_{69}	−95715
c_4	10	c_{26}	−4027775	c_{48}	347890288	c_{70}	−3419626
c_5	−44	c_{27}	3400128	c_{49}	380804382	c_{71}	279592
c_6	317	c_{28}	34694977	c_{50}	222958790	c_{72}	−131688
c_7	−441	c_{29}	65080327	c_{51}	247189038	c_{73}	−159836
c_8	−380	c_{30}	57521147	c_{52}	156335657	c_{74}	−129246
c_9	117	c_{31}	59618093	c_{53}	−62117809	c_{75}	79954
c_{10}	−3592	c_{32}	61117524	c_{54}	68720838	c_{76}	1844
c_{11}	−2826	c_{33}	9242717	c_{55}	72198432	c_{77}	−24819
c_{12}	−4150	c_{34}	−53405172	c_{56}	−67785311	c_{78}	−3511
c_{13}	−15878	c_{35}	−200240057	c_{57}	−29371390	c_{79}	14207
c_{14}	30770	c_{36}	−432469442	c_{58}	−125917150	c_{80}	5717
c_{15}	59701	c_{37}	−445800343	c_{59}	−202263007	c_{81}	−368
c_{16}	105016	c_{38}	−311597166	c_{60}	−95170057	c_{82}	−249
c_{17}	242607	c_{39}	−280198147	c_{61}	−45125822	c_{83}	−27
c_{18}	217600	c_{40}	−137857186	c_{62}	−6797306	c_{84}	22
c_{19}	136672	c_{41}	20973696	c_{63}	27946797	c_{85}	−12
c_{20}	−46108	c_{42}	141344229	c_{64}	6623766		
c_{21}	−1445658	c_{43}	435553059	c_{65}	21795014		
c_{22}	−2751376	c_{44}	593203246	c_{66}	7080040		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 85, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $85 + 4$ terms removed returns order 85 as well, so $\deg q_{\min} = 85 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $85 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 85 that this sequence satisfies — and that family turns out to have exactly one member.

References

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